

ADDER CIRCUIT

Condensed Study Notes

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Half Adders

A half adder is a digital circuit that adds two single binary bits. It produces two outputs: a Sum (S) bit and a Carry (C) bit.

There are four possible combinations when adding two bits (A and B):

- $0 + 0 = 0$ (S=0, C=0)
- $0 + 1 = 1$ (S=1, C=0)
- $1 + 0 = 1$ (S=1, C=0)
- $1 + 1 = 10$ (S=0, C=1) (The '10' represents a sum of 0 in the ones column and a carry of 1 to the next column)
- **Sum (S):** The Sum bit is 1 only when the two input bits (A and B) are different. This behavior is identical to an XOR (exclusive OR) operation. The Sum is also described as the modulo-2 sum of the addends, meaning the result of addition where you 'loop around' to zero after passing one.
- **Carry (C):** The Carry bit is 1 only when both input bits (A and B) are 1. This behavior is identical to an AND operation.

COMPUTER ARCHITECTURE ADDER CIRCUIT

An adder is a circuit that sums two bits, called addends, and provides a two-bit output. Adders can be linked together to sum binary numbers of any length.

There are two main types of adders:

- **Half Adder:** Adds two single bits.
- **Full Adder:** Adds two bits plus a carry-in bit from a previous addition.

Full Adders

Full adders address the limitation of half adders by incorporating a carry-in input, allowing them to sum three one-bit numbers: the two addends and a carry-in (Cin).

This enables full adders to handle carry signals, making them suitable for adding multiple bits together, unlike stand-alone half adders.

A full adder can be built using two half adders. The first half adder takes the two addends (A and B). Its sum output is fed into the second half adder along with the carry-in (Cin). The carry outputs from both half adders are then ORed together to produce the final carry-out (Cout).

The output of a full adder represents the two-bit arithmetic sum of the three input bits.

Reference

The provided content for the 'Reference' section is sparse and does not contain enough information to generate condensed study notes or a summary. It appears to be a placeholder or a link that does not directly provide translatable text for summarization.